7.1 Integration by Parts

Recall the Product Rule.

$$\int \frac{d}{dx}[f(x)g(x)] = \int f(x)\ g'(x) \ + \int g(x)\ f'(x)$$

We can undo this process with an integral to give us the following (Note: We will worry about +C later):

$$f(x)g(x) = \int f(x)\ g'(x) + \int f'(x)\ g(x)$$

Rearranging this gives us the following:

$$\int f(x)g'(x) = f(x)g(x) - \int f'(x)g(x)$$

Application of the above (or below) formula(s) is called **Integration by Parts**. Most of the time we remember this formula using the following notation. If $u = f(x)$ and $v = g(x)$, then using differential notation, $du = f'(x)\ dx$ and $dv = g'(x)\ dx$, and we have:

$$\boxed{\int u\ dv \ = \ uv - \int v\ du}$$

The goal is to replace a difficult integral with something simpler, hopefully a Day 1 Integral.

Integrals of the general form $\int x^m e^x\ dx$

u = derivation gets simpler
dv = integral gets simpler

1. (a) $\int_0^1 xe^{3x}\ dx$

$u = x \quad dv = e^{3x}$

$du = 1 \quad v = \frac{e^{3x}}{3}$

$\frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} \Rightarrow \left[\frac{1}{3}e^3 - \frac{1}{9}e^3\right] - \left[-\frac{1}{9}\right] = \frac{2}{9}e^3 + \frac{1}{9}$

$\hookrightarrow \frac{xe^{3x}}{3} - \int \frac{e^{3x}}{3} \Big|_0^1$

(b) $\int (x^2+5)e^{-2x}\ dx$

$u = x^2+5 \quad dv = e^{-2x}$

$du = 2x \quad v = -\frac{1}{2} \cdot e^{-2x}$

$(x^2+5)\left(-\frac{1}{2}e^{-2x}\right) - \int -\frac{1}{2}e^{-2x}\ 2x\ dx$

$(x^2+5)\left(-\frac{1}{2}e^{-2x}\right) - \left[2x \cdot -\frac{1}{4}e^{-2x} - \int +\frac{1}{4}e^{-2x} \cdot 2\right]$

$(x^2+5)\left(-\frac{1}{2}e^{-2x}\right) + \frac{1}{2}xe^{-2x} + -\frac{1}{4}e^{-2x} + C$

$a = 2 \quad db = \frac{1}{4}e^{-2x}$

$da = 0 \quad b = -\frac{1}{8}e^{-2x}$

$z = 2x \quad dw = -\frac{1}{2}e^{-2x}$

$dz = 2 \quad w = +\frac{1}{4}e^{-2x}$

(c) $\int x^7 e^{-x^4}\, dx$

$u = -x^4$

$du = -4x^3\, dx$

$\frac{1}{4} du = x^3 dx$

Tabular

$z = u$	$+$	$dv = e^u$
1	$-$	e^u
0	$+$	e^u

$\int x^4 \cdot x^3 e^{-x^4}\, dx$

$\frac{1}{4}\int (-u)(e^u)\, du = ue^u - e^u + C$

$\frac{1}{4}\left[-e^u e^{-x^4} \cdot e^{-x^4}\right]$

Integrals of the general form $\int x^m \cos x\, dx$ or $\int x^m \sin x\, dx$

2. (a) $\int 3x \sin 2x\, dx$

$u = 3x \quad dv = \sin 2x$

$du = 3 \quad v = -\frac{1}{2}\cos(2x)$

$3x \cdot -\frac{1}{2}\cos(2x) - \int -\frac{3}{2}\cos(2x)$

$-\frac{3}{2}\cos(2x)x + \frac{3}{2}\int \cos(2x)$

$-\frac{3}{2}\cos(2x)x + \frac{3}{2}\cdot\frac{1}{2}\sin(2x) + C$

(b) $\int 2x^5 \cos(x^2)\, dx$

u-sub

$u = x^2$

$du = 2x\, dx$

$\frac{1}{2} du = x\, dx$

$\frac{2}{2}\int u^2 \cos(u)\, du$

Tabular

$z = u^2$	$+$	$dv = \cos(u)$
$dz = 2u$	$-$	$\sin(u)$
2	$+$	$-\cos(u)$
0	$-\int$	$-\sin(u)$

$= u^2 \sin u + 2u\cos u - 2\sin u + C$

$$\boxed{x^4 \sin x^2 + 2x^2 \cos x^2 - 2\sin x^2 + C}$$

Integrals of the general form $\int x^m \ln x \, dx$

3. (a) $\int x^7 \ln x \, dx$

$u = \ln x \qquad dv = x^7$

$du = \frac{1}{x} \qquad v = \frac{x^8}{8}$

$= \ln x \, \frac{1}{8} x^8 - \int \frac{1}{x} \frac{x^8}{8} dx$

$= \frac{1}{8} x^8 \ln x - \frac{1}{8} \int x^7 dx$

$= \frac{1}{8} x^8 \ln x - \frac{1}{8} \frac{x^8}{8} + C \qquad = \frac{1}{8} x^8 \ln x - \frac{x^8}{64} + C$

(b) $\int \frac{\ln x}{x^2} \, dx$

$u = \ln x \qquad dv = x^{-2}$

$du = \frac{1}{x} \qquad v = -\frac{1}{x}$

$\ln x \left(-\frac{1}{x}\right) + \int x^{-2} dx$

$\frac{-\ln(x)}{x} - \frac{1}{x} + C$

(c) $\int \ln x \, dx$

$u = \ln x \qquad dv = dx$

$du = \frac{1}{x} \qquad v = x$

$= \ln x \, (x) - \int 1 \, dx$

$= x \ln x - x + C$

Other types: Inverse Trig and Loops

4. $\int \arcsin(x)\ dx$

$u = \sin^{-1}(x) \qquad dv = dx$

$du = \frac{1}{\sqrt{1-x^2}}\,dx \qquad v = x$

$\sin^{-1}(x)x - \int \frac{x}{\sqrt{1-x^2}}\,dx$

u-sub: $z = 1-x^2$

$dz = -2x\,dx$

$-\frac{1}{2}dz = x\,dx$

$\sin^{-1}(x)x + \frac{1}{2}\int z^{-1/2}\,dz$

$x\sin^{-1}(x) + \frac{1}{2}(2)\,z^{1/2} + C$

$x\sin^{-1}(x) + \sqrt{z} + C \implies x\sin^{-1}(x) + \sqrt{1-x^2} + C$

5. $\int e^x \cos 2x\ dx$

$u = \cos(2x) \qquad dv = e^x$

$du = -2\sin(2x)\,dx \qquad v = e^x$

$\cos(2x)e^x + \int e^x 2\sin 2x\,dx$

$z = 2\sin 2x \qquad dw = e^x$

$dz = 4\cos(2x) \qquad w = e^x$

$= e^x\cos 2x + 2\sin(2x)e^x - 4\int e^x\cos(2x)$

$5\int e^x\cos(2x) = e^x\cos(2x) + 2\sin(2x)e^x$

$\int e^x\cos(2x) = \frac{e^x\cos(2x) + 2e^x\sin(2x)}{5} + C$

Another type of integral that would require a loop would be $\int \cos(3x)\sin(2x)\ dx$.

7.2 Trig Integrals

Identities for this section:

$\sin^2 x + \cos^2 x = 1$ which means that $\sin^2 x = 1 - \cos^2 x$ and $\cos^2 x = 1 - \sin^2 x$

$\tan^2 x + 1 = \sec^2 x$ which means that $\tan^2 x = \sec^2 x - 1$

$\cot^2 x + 1 = \csc^2 x$ which means that $\cot^2 x = \csc^2 x - 1$

$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

Most Trig Integrals are really just substitutions. The "technique" here is just understanding the thought process when we see high powers of trig functions. The goal is to focus on the potential derivative term, du, for different choices of trig functions for u. After we factor out du, we want to be left with an expression we can rewrite with trig identities. Most of the time this is achieved by making sure whatever non-u terms remain involve even exponents.

1. (a) $\int \sin^3 x \cos^6 x \, dx$

$u = \cos(x)$

$du = -\sin dx$

$-1 du = \sin(x) dx$

$-\int (1-\cos^2 x)\cos^6 x \, dx$

$-\int (1-u^2) u^6 du = -\int u^6 - u^8 = -\frac{1}{7}u^7 + \frac{1}{9}u^9 + C$

$= -\frac{1}{7}(\cos^7 x) + \frac{1}{9}\cos^9 x + C$

(b) $\int \sin^6 x \cos^5 x \, dx$

$u = \sin x$

$du = \cos x$

$\int \sin^6 x \cos^4 x \cos x \, dx$

$(\cos^2 x)^2$

$\int u^6 (1-\sin^2 x)^2 du$

$= \int u^6 (1-u^2)^2 du$

$= \int u^6 (1 + u^4 - 2u^2)$

$\Rightarrow \int u^6 + u^{10} - 2u^8$

$\Rightarrow \frac{u^7}{7} + \frac{u^{11}}{11} - \frac{2u^9}{9}$

$\frac{\sin^7 x}{7} + \frac{\sin^{11} x}{11} - \frac{2\sin^9 x}{9} + C$

(c) $\int \cos^3 x \, dx$

$\cos^2 x \cos x \, dx$

$u = \sin x$

$du = \cos x \, dx$

$\int (1 - \sin^2 x) \cos x \, dx$

$\int (1 - u^2) \, du$

$= u - \frac{u^3}{3}$

$= \sin x - \frac{\sin^3 x}{3}$

(d) $\int \sin(5x) \tan^2(5x) \, dx$

$\int \sin(5x) \frac{\sin^2(5x)}{\cos^2(5x)} \, dx$

$u = \cos(5x)$

$du = -\sin(5x) \cdot 5 \, dx$

$-\frac{1}{5} du = \sin(5x)$

$= \int \frac{\sin^3(5x)}{\cos^2(5x)} \, dx = -\frac{1}{5} \int \frac{\sin^2(5x) \sin(5x)}{u^2}$

$= -\frac{1}{5} \int \frac{(1 - \cos^2(5x)) \sin(5x)}{u^2}$

$= -\frac{1}{5} \int \frac{(1 - u^2)}{u^2} \, du = -\frac{1}{5} \int u^{-2} - 1$

$-\frac{1}{5}(-u^{-1} - u) + C$

$-\frac{1}{5}\left(-\frac{1}{\cos(5x)} - \cos(5x)\right) + C$

(e) $\int \sin^7 x \cos^3 x \, dx$

$u = \sin x$

$du = \cos x$

$\int u^7 (1 - u^2) \, du$

$\tan^2 x + 1 = \sec^2 x$

(SL sec DE)

2. (a) $\int \tan^2 x \, dx$

$\int \sec^2 x - 1 \, dx = \tan x - x + C$

(b) $\int \tan^8 x \sec^4 x \, dx$

$u = \tan x$

$du = \sec^2 x$

$\int \tan^8 x \sec^2 x \sec^2 x \, dx$

$\int \tan^8 x (\tan^2 x + 1) \sec^2 x \, dx$

$\int u^8 (u^2 + 1) \, du \Rightarrow \int u^{10} + u^8$

(c) $\int x \tan^5(x^2) \sec^3(x^2) \, dx$

$u = \sec(x^2)$

$du = \sec(x^2) \tan(x^2) \cdot 2x \, dx$

$\frac{1}{2} du = \sec(x^2) \tan(x^2) \cdot x \, dx$

$\int x \tan^4(x^2) \sec^2(x^2) \sec(x^2) \tan(x^2) \, dx$

$\frac{1}{2} \int (\sec^2 - 1)^2 u^2 \, du$

$\frac{1}{2} \int (u^2 - 1)^2 u^2 \, du$

(d) $\int \sec^4 x \, dx$

$u = \tan x$

$du = \sec^2 x \, dx$

$\int \sec^2 x \sec^2 x$

$\int (\tan^2 x + 1) \sec^2 x \, dx$

$\int (u^2 + 1) \, du$

$\cot^2 x + 1 = \csc^2 x$

(e) $\int \cot^3 x \csc x \, dx$

$u = \csc x$

$du = -\csc x \cot x$

$-du = \csc x \cot x$

$\int \cot^2 x \cot x \csc x \, dx$

$-\int (\csc^2 x - 1) \, du$

$-\int (u^2 - 1) \, du$

3. (a) $\int \sin^2 x \, dx = \int \frac{1}{2} [1 - \cos(2x)] \, dx$

$= \frac{1}{2} [x - \frac{1}{2} \sin(2x)] + C$

(b) $\int \sin^2 x \cos^2 x \, dx = \int \frac{1}{2}(1 - \cos(2x)) \frac{1}{2}(1 + \cos(2x)) \, dx$

$\frac{1}{4} \int (1 - \cos^2(2x)) \, dx = \frac{1}{4} \int \sin^2 x \, dx$

$\frac{1}{4} \int \frac{1}{2} [1 - \cos(4x)]$

$\frac{1}{8} (x - \frac{1}{4} \sin(4x)) + C$

4. $\int \sec x \, dx$